NARAYANA ENGINEERING COLLEGE :: GUDUR

DEPARTMENT OF ECE

CourseDetails							
Class	:	**II -B.Tech**	**Semester**	**II**	**Year**	:	**2024-25**
Course Title	:	**EMTL**	**Course Code**	: **23EC2005**	**Credits**	:	**3**
Program/Dept.	:	**ECE**	**Section**	: **A, B, C, EVT & ECA**	**Batch**	:	**2023- 2027**
Regulation	:	**NECR-23**	**Faculty**	**:Dr. B. Malakonda Reddy & Mrs.P.Manjulamma**			

Question Bank /Topic Learning Outcomes

(Table format as given below is suitable for Theory, MCQ, Lab, and Tutorial. Columns can be added or deleted. No need to take printouts)

TLO. No:	Module No:	Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	CO	BL	Marks per Q
MODULE – 1 ELECTROSTATICS						
1	1	1	Define electric field intensity. Obtain an expression for the electric field intensity at a point which is at a distance of R from a point charge Q. **ANS:** Consider a point charge Q_1 As we know, if any other similar charge Q_2 is brought near it, Q_2 experiences a force	1	2	5

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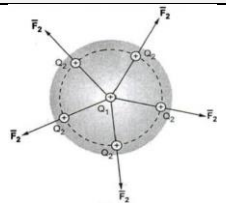
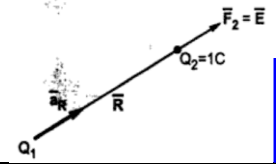
			 If Q_2 is moved around Q_1, still Q_2 experiences a force. The region where a particular charge exerts a force on other charge located in that region is called electric field of that charge. Based on Coulomb's Law force on Q_2 due to Q_1 $$\bar{F}_2 = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R_{12}^2}\bar{a}_{12}$$ Thus force per charge can be written as $$\frac{\bar{F}_2}{Q_2} = \frac{Q_1}{4\pi\varepsilon_0 R_{12}^2}\bar{a}_{12}$$ From this, '*Force per unit charge*' is called as **Electric Field Intensity** or **Electric Field Strength.** $$\bar{E} = \frac{Q_1}{4\pi\varepsilon_0 R_{1p}^2}\bar{a}_{1p}$$ where p = position any other charge (unit charge) around Q_1  $$\bar{E} = \frac{Q_1}{4\pi\varepsilon_0 R^2}\bar{a}_R$$			

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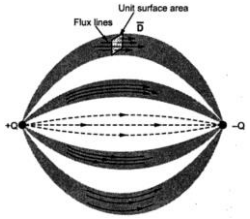
			If charge Q is located at the centre of the sphere which is having radius 'r' then the electric field intensity on the surface of the sphere wall be written as $$\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\bar{a}_r$$ Units of The definition of Electric Field Intensity $\bar{E}$ is $$\bar{E} = \frac{\text{Force}}{\text{Unit charge}} = \frac{(N)\ \text{Newtons}}{(C)\ \text{Coulomb}}$$ Hence the units of $\bar{E}$ is N/C ; But in practically it has Volts/metre from **E = V/d**			
			Define electric flux density. Derive the relation between $\bar{E}$ and $\bar{D}$. ANS: Consider unit surface area, the number of flux lines are passing through this surface area. The net flux passing normal through the unit surface area is called **Electric Flux Density.**  It has specific direction which is normal to the surface, hence it will represent as a Vector. It is denoted with $\bar{D}$. The flux density vector is defined to be independent of the material media. Consider a sphere with charge Q at its centre. The total flux(ψ = Q) is passing normally through total surface area (S).	1	2	5

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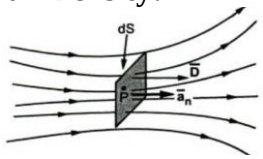
			Then the flux density is defined as $D = \frac{\psi}{S}$ in magnitude Units are **Coulombs/m^2 .** Electric Flux Density is also called as **Displacement Flux Density** or **Displacement Density.** **Vector form of Electric Flux Density:**  Consider the differential surface area dS at point 'P'. The flux passing through dS is dψ. The direction of D is same as flux lines i.e., normal to the surface area i.e., $\bar{a}_n$ (unit vector normal to the surface area). Hence the Flux Density $\bar{D}$ at point P can be represented in vector form as $\bar{D} = \frac{d\psi}{dS}\bar{a}_n$ C/m^2 where $\bar{a}_n =$ unit vector normal to the differential surface area **Relation between $\bar{D}$ and $\bar{E}$** Electric Field Intensity at a point which is at a distance of 'r' from the point charge Q is given by $\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\bar{a}_r$ Electric Flux Density at a point which is at a distance of 'r' from the point charge Q is given by $\bar{D} = \frac{Q}{4\pi r^2}\bar{a}_r$			

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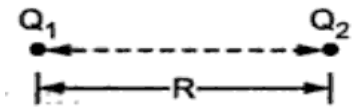
			Dividing the Eqs $\overline{D}$ **and** $\overline{E}$ $$\frac{\overline{D}}{\overline{E}} = \frac{\frac{Q}{4\pi\varepsilon_0 r^2}\overline{a}_r}{\frac{Q}{4\pi r^2}\overline{a}_r} = \varepsilon_0$$ $\overline{D} = \varepsilon_0 \overline{E}$ for free space $\overline{D}$ and $\overline{E}$ related through permittivity. If the medium is considered other than free space $\overline{D} = \varepsilon_0 \varepsilon_r \overline{E}$; ε_r is relative permittivity $\overline{D} = \varepsilon \overline{E}$			
2	1	2	State Coulomb's law and derive expression for force between two point charges. ANS: **Coulomb's Law** states that the force between two point charges $Q_1$and Q_2 ➢ acts along the line joining the two point charges ➢ is directly proportional to the product (Q_1Q_2) of the two charges ➢ is inversely proportional to the square of the distance between them. Let us consider, two point charges $Q_1$and Q_2 separated by the distance 'R'  The charge Q_1 apply a force on Q_2 , while the Q_2 also apply a force on Q_1 Mathematically **the force F** between the charges can be expressed as	1	2	10

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			$$F \propto \frac{Q_1Q_2}{R^2}$$ where Q_1Q_2 = Product of the two charges **R** = Distance between the two charges Proportionality constant is introduced by eliminating proportionality in the above equation $$F = k\frac{Q_1Q_2}{R^2}$$ where **k** is the proportionality constant. To satisfy the coulomb's law, the proportionality constant is defined as $$k = \frac{1}{4\pi\varepsilon} \qquad F = \frac{1}{4\pi\varepsilon}\frac{Q_1Q_2}{R^2}$$ where **ε** is permittivity of the medium in which charges are located In general **ε** is expressed as $$\varepsilon = \varepsilon_0\varepsilon_r$$ where ε_0 = Permittivity of free space or Vacuum ε_r = Relative Permittivity or Dielectric constant of the medium **ε** = Absolute Permittivity The units of **ε** are farads/metre (F/m) For free space or Vacuum the Relative Permittivity $\varepsilon_r = 1$; Hence $\varepsilon = \varepsilon_0$ $$F = \frac{1}{4\pi\varepsilon_0}\frac{Q_1Q_2}{R^2}$$ the value of Permittivity of free space ε_0 is $$\varepsilon_0 = \frac{1}{36\pi}\times 10^{-9} = 8.854\times 10^{-12}\ F/m$$ $$k = \frac{1}{4\pi\varepsilon_0} = \frac{1}{4\pi\times 8.854\times 10^{-12}} = 8.98\times 10^9 \approx 9\times 10^9\ m/F$$			

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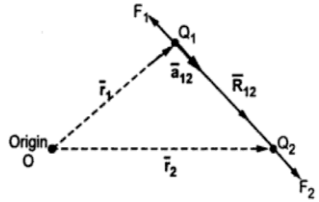
			If the two charges are located in free space, then the coulomb's law gives the force between the charges as $$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}$$ **Vector Representation of Coulomb's Law**  If the two charges Q_1 and Q_2 are located at the points having the position vectors $\overline{r_1}$ and $\overline{r_2}$ The force on Q_2 due to Q_1 acts along the line R_{12} or along the distance vector $\overline{R}_{12}$. where $\overline{a}_{12}$ is unit vector along $\overline{R}_{12}$; it represents the direction of $\overline{R}_{12}$ The force on Q_2 due to Q_1 can be expressed in vector form as $$\bar{F}_2 = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R_{12}^2}\bar{a}_{12} \quad ; \quad \bar{a}_{12} = \frac{\bar{R}_{12}}{\lvert\bar{R}_{12}\rvert} = \frac{\bar{r}_2 - \bar{r}_1}{\lvert\bar{R}_{12}\rvert} = \frac{\bar{r}_2 - \bar{r}_1}{\lvert\bar{r}_2 - \bar{r}_1\rvert}$$ Where $\lvert\overline{R}_{12}\rvert$ = R = Distance between two charges **OBSERVATIONS:** 1. The force applied by two charges on each other is equal but opposite in direction.			

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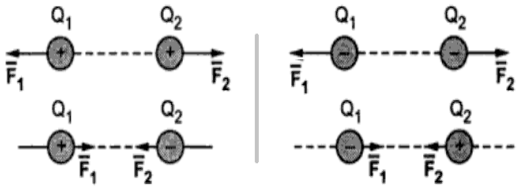
			$$\bar{F}_1 = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R_{21}^2}(-\bar{a}_{21}) = -\bar{F}_2$$ 2. The like charges (same polarity charges) repel each other The unlike charges (opposite polarity charges) attract each other  3. It is necessary that the two charges are point charges and stationary in nature. 4. The Coulomb's law is linear, which shows that, if any one charge is increased 'n' times then the force exerted also increases by 'n' times $$\bar{F}_2 = -\bar{F}_1 \quad \text{then} \quad n\bar{F}_2 = -n\bar{F}_1$$ **n** : scalar			
3	1	3	Find the electric field strength due to Point and infinite sheet of charge or infinite charged surface area ANS: **Electric field Intensity due to Point Charge** If charge Q is located at the centre of the sphere which is having radius 'r' then the electric field intensity on the surface of the sphere wall be written as	1	2	10

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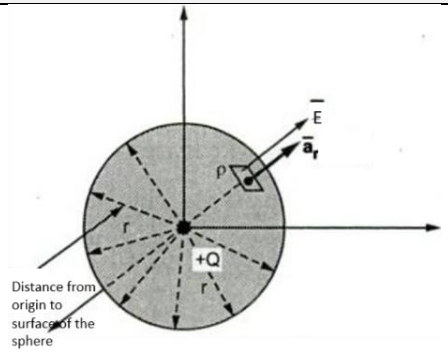
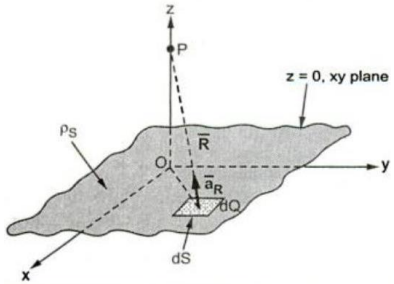
			 $\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\bar{a}_r$ **Electric field Intensity due to Infinite Surface Charge** Consider a Infinite sheet of charge with charge density ρ_S C/m^2 on xy-plane (i.e., z = 0 plane) which is from -∞ to +∞ in Cylindrical coordinate system. Due to this Infinite surface charge Electric field Intensity has to be calculated at point P on z-axis which is 'z' distance from origin.  Consider a differential surface area **dS** carrying a charge ***dQ*** over infinite surface $dQ = \rho_S dS$			

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			Since the differential surface area **dS** is normal to z-axis **dS = rdrdϕ** $dQ = r\rho_S drd\phi$ Hence the differential electric field $d\overline{E}$ at point P due to dQ is given by $d\overline{E} = \frac{dQ}{4\pi\varepsilon_0 R^2}\overline{a}_R = \frac{\rho_S dS}{4\pi\varepsilon_0 R^2}\overline{a}_R$ $\overline{a}_R = \frac{\overline{R}}{\left\lvert\overline{R}\right\rvert}$; $\overline{R} = \overline{1P} = \overline{P} - \overline{1} = z\overline{a}_z - r\overline{a}_r$; $\left\lvert\overline{R}\right\rvert = \sqrt{r^2 + z^2}$ $\overline{a}_R = \frac{\overline{R}}{\left\lvert\overline{R}\right\rvert} = \frac{z\overline{a}_z - r\overline{a}_r}{\sqrt{r^2 + z^2}}$ $d\overline{E} = \frac{\rho_S dS}{4\pi\varepsilon_0 R^2}\overline{a}_R = \frac{r\rho_S drd\phi}{4\pi\varepsilon_0(\sqrt{r^2 + z^2})^2}\frac{z\overline{a}_z - r\overline{a}_r}{\sqrt{r^2 + z^2}}$ Here charge is on xy-plane; $\overline{a}_r \rightarrow \text{parallel} \rightarrow \text{xy} - \text{plane}$; $\overline{a}_r \rightarrow \text{parallel} \rightarrow ch\arg e$ Because of this, in $d\overline{E}$, $\overline{a}_r$ component will be cancelled or neglected. Hence the equation of $d\overline{E}$ can be written by eliminating the r-component $d\overline{E} = \frac{\rho_S dS}{4\pi\varepsilon_0 R^2}\overline{a}_R = \frac{r\rho_S drd\phi}{4\pi\varepsilon_0(\sqrt{r^2 + z^2})^2}\frac{z\overline{a}_z}{\sqrt{r^2 + z^2}}$ Now by integrating $d\overline{E}$ over **dS = rdrdϕ**, total $\overline{E}$ will be obtained at point 'P'			

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			• r → limits → 0 to ∞ • ϕ → limits → 0 to 2π $\bar{E} = \int_{\phi=0}^{2\pi}\int_{r=0}^{\infty} d\bar{E} = \int_{0}^{2\pi}\int_{0}^{\infty} \frac{\rho_S\, r\, dr\, d\phi}{4\pi\varepsilon_0 (r^2+z^2)^{3/2}} (z\bar{a}_z)$ To solve this, put **$r^2 + z^2 = u^2$** → **2rdr = 2udu** → **rdr = udu** for **r = 0** → **u = z** ; **r = ∞** → **u = ∞** $\bar{E} = \int_{0}^{2\pi}\int_{u=z}^{\infty} \frac{\rho_S}{4\pi\varepsilon_0}\frac{udu}{(u^2)^{3/2}}\, d\phi\; z\,\bar{a}_z$ $\bar{E} = \int_{0}^{2\pi}\int_{u=z}^{\infty} \frac{\rho_S}{4\pi\varepsilon_0}\frac{udu}{(u^2)^{3/2}}\, d\phi\; z\,\bar{a}_z$ $= \int_{0}^{2\pi}\int_{u=z}^{\infty} \frac{\rho_S}{4\pi\varepsilon_0}\frac{du}{u^2}\, d\phi\; (z\,\bar{a}_z)$ $= \int_{0}^{2\pi} \frac{\rho_S}{4\pi\varepsilon_0}\, d\phi\; z\,\bar{a}_z \left[-\frac{1}{u}\right]_z^{\infty}$ $= \frac{\rho_S}{4\pi\varepsilon_0}[\phi]_0^{2\pi}\,(z\bar{a}_z)\left[-\frac{1}{\infty}-\left(-\frac{1}{z}\right)\right] = \frac{\rho_S}{4\pi\varepsilon_0}(2\pi)\,\bar{a}_z$ $\bar{E} = \frac{\rho_S}{2\varepsilon_0}\,\bar{a}_z \;\; V/m$ $\bar{a}_z$ direction is normal to differential surface area **dS**.			

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			Hence in general if $\bar{a}_n$ is direction normal to the surface containing charge, $\bar{E}$ can be written as $\bar{E} = \frac{\rho_s}{2\varepsilon_0}\bar{a}_n \quad V/m$ $\bar{a}_n =$ normal unit vector to the surface For the points below xy-plane $\bar{a}_n = -\bar{a}_z$ $\bar{E} = -\frac{\rho_s}{2\varepsilon_0}\bar{a}_z \quad V/m$			
4	1	4	Derive the Maxwell's equations for electrostatics. ANS: **Maxwell's First Equation for Electrostatics** The divergence of $\bar{D}$ is given by $div\bar{D} = Lt_{\Delta v \to 0}\frac{\oint_s \bar{D} \bullet d\bar{S}}{\Delta v} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$ According to Gauss's Law $\psi = Q = \oint_s \bar{D} \bullet d\bar{S}$ Express the Gauss's Law per unit volume basis $\frac{Q}{\Delta v} = \frac{\oint_s \bar{D} \bullet d\bar{S}}{\Delta v}$	1	2	5

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			Take both sides $Lt_{\Delta v \to 0}$ $$Lt_{\Delta v \to 0} \frac{Q}{\Delta v} = Lt_{\Delta v \to 0} \frac{\oint_s \overline{D} \bullet d\overline{S}}{\Delta v}$$ But $$Lt_{\Delta v \to 0} \frac{Q}{\Delta v} = \rho_v$$ and $$Lt_{\Delta v \to 0} \frac{\oint_s \overline{D} \bullet d\overline{S}}{\Delta v} = div\overline{D}$$ Now $$div\overline{D} = \rho_v \qquad \textbf{i.e.,} \qquad \nabla \bullet \overline{D} = \rho_v$$ Finally this equation $$\nabla \bullet \overline{D} = \rho_v$$ called Maxwell's first equation applied to electrostatics. This is also called as point form of Gauss's Law or Gauss's Law in differential Form. The statement of Gauss's Law in point Form is "*The divergence of electric flux density in the medium at a point is equal to the volume charge density at the same point*"			

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			Maxwell's Second Equation for Electrostatics The work done in moving a test charge around any closed path in static electric filed is **ZERO**. Here starting and terminating points are same for a closed path; hence the upper and lower limits of integration becomes same; hence work done becomes **ZERO**. $$\oint_L \overline{E} \bullet d\overline{L} = 0$$ Whenever the amount of work done is zero then this property is called as "*Conservative property of the field or lamellar property of the field*". Now based on Conservative property of the field, we can write $$\oint_L \overline{E} \bullet d\overline{L} = 0$$ This is the Maxwell's Second Equation for electrostatics in integral form. Apply stroke's theorem L.H.S for the above equation $$\oint_L \overline{E} \bullet d\overline{L} = \int_s (\nabla X \overline{E}) \bullet d\overline{S} = 0$$ We will get $$\nabla X \overline{E} = 0$$ This is the Maxwell's Second Equation for electrostatics in point form or differential form.			

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			Given the electric flux density $\vec{D} = e^{-x}\sin(y)\,a_x - e^{-x}\cos(y)\,a_y + za_z$ C/m^2, determine an approximate value for the total charge enclosed in an incremental volume of 10^{-9}m^3 located at the origin. **ANS:** $$\vec{D} = e^{-x}\sin(y)\,a_x - e^{-x}\cos(y)\,a_y + za_z$$ To find the total charge enclosed within an incremental volume of 10^{-9} m^3 located at the origin. we'll use the divergence of the electric flux density D to determine the charge density ρ, and then integrate it over the given volume. Divergence of flux density is equal to volume charge density . i.e., $$\nabla \cdot \bar{D} = \rho_v$$ $$\nabla \cdot \bar{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \frac{\partial e^{-x}\sin(y)}{\partial x} + \frac{\partial(-e^{-x}\cos(y))}{\partial y} + \frac{\partial z}{\partial z}$$ $$\nabla \cdot \bar{D} = -\sin(y)\,e^{-x} + e^{-x}\sin(y) + 1 = 1$$ The charge density within the given volume is constant and equal to 1C/m^3 Now, to find the total charge enclosed within the volume, multiply the charge density by the volume: Given volume = 10^{-9} m^3 **Total Charge = Charge density X Volume = 1 X 10^{-9} = 10^{-9} C**	1	2	5

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5	1	5	Four positive charges10^{-9}C each are situated in x-y plane at points (0,0), (0,1), (1,1) and (1,0) m.finds the electric field intensity and electric potential at (1/2, 1/2). ANS: Given data $Q_1 = 10^{-9}$ C at $P_1 = (0,0)$m $Q_2 = 10^{-9}$ C at $P_2 = (0,1)$m $Q_3 = 10^{-9}$ C at $P_3 = (1,1)$m $Q_4 = 10^{-9}$ C at $P_4 = (1,0)$m At $P_5 = (1/2, 1/2)$m, we have to find E and V $R_1 = P_1P_5 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(1/2 - 0)^2 + (1/2 - 0)^2} = 1/\sqrt{2}m$ $R_2 = P_2P_5$ $R_3 = P_3P_5$ $R_4 = P_4P_5$ $E_1 = \frac{Q_1}{4\pi\varepsilon_0 R_1^2}$ $E_2 = \frac{Q_2}{4\pi\varepsilon_0 R_2^2}$ $E_3 = \frac{Q_3}{4\pi\varepsilon_0 R_3^2}$ $E_4 = \frac{Q_4}{4\pi\varepsilon_0 R_4^2}$ $E = E_1 + E_2 + E_3 + E_4$ $V_1 = \frac{Q_1}{4\pi\varepsilon_0 R_1}$ $V_2 = \frac{Q_2}{4\pi\varepsilon_0 R_2}$ $V_3 = \frac{Q_3}{4\pi\varepsilon_0 R_3}$ $V_4 = \frac{Q_4}{4\pi\varepsilon_0 R_4}$ $V = V_1 + V_2 + V_3 + V_4$	1	2	5

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			Derive the Poison's and Laplacian Equations in electrostatics ANS: **POISSON'S EQUATION** Poisson's Equation derived from Gauss's Law Consider the point form of Gauss's Law $\nabla \cdot \overline{D} = \rho_v$ For a homogeneous, isotropic and linear medium, the flux density and electric field intensity are directly proportional. Thus, $\overline{D} = \varepsilon \overline{E}$ $\nabla \cdot \overline{D} = \rho_v$ $\nabla \cdot \varepsilon \overline{E} = \rho_v$ From the gradient relationship $\overline{E} = -\nabla V$ $\nabla \cdot \varepsilon \overline{E} = \rho_v$ $\nabla \cdot \varepsilon(-\nabla V) = \rho_v$ $-\varepsilon[\nabla \cdot \nabla V] = \rho_v$ $\nabla \cdot \nabla V = -\frac{\rho_v}{\varepsilon}$	1	2	5

TLO. No:	Module No:	Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	CO	BL	Marks per Q
			Now ∇.∇ operation is called 'del squared' operation and denoted as ∇^2 Therefore $$\nabla \cdot \nabla V = -\frac{\rho_v}{\varepsilon}$$ $$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$$ This equation is Poisson's Equation **LAPLACE'S EQUATION** In a certain region, for dielectric medium, volume charge density is zero ($\rho_v = 0$), then Poisson's equation takes the form $$\nabla^2 V = 0$$ (for charge free region) This is called Laplace's Equation The ∇^2 operation is called as Laplacian of V In Cartesian Co-ordinate System $$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$ In Cylindrical Co-ordinate System $$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\left(\frac{\partial^2 V}{\partial \phi^2}\right) + \frac{\partial^2 V}{\partial z^2} = 0$$ In Spherical Co-ordinate System $$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2} = 0$$			

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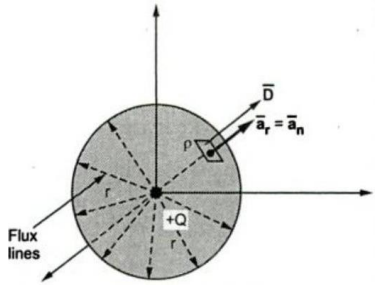
6	1	6	Obtain an expression for electric field intensity using Gauss law. ANS: $\overline{E}$ **due to a point charge Q** Consider a sphere with charge Q at its centre.  Sphere radius is 'r' Flux lines are passing normal to the surface, i.e., in the direction of $\overline{a}_n = \overline{a}_r$ The magnitude of $\overline{D}$ is $\lvert\overline{D}\rvert = \frac{Totalflux(\psi)}{TotalArea(S)}$ but Total Flux = ψ = Q Total Surface area = S = $4\pi r^2$ $$\lvert\overline{D}\rvert = \frac{Totalflux(\psi)}{TotalArea(S)}$$ $$\lvert\overline{D}\rvert = \frac{Q}{4\pi r^2}$$ Since the flux lines direction is $\overline{a}_n = \overline{a}_r$; Flux Density is also having the same direction.	1	1	5

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			Finally, Electric Flux Density at a point which is at a distance of 'r' from the point charge Q is given by $\overline{D} = \frac{Q}{4\pi r^2}\overline{a}_r C/m^2$ The flux density vector is defined to be independent of the material media Now from the relation between $\overline{D}$ and $\overline{E}$ $\overline{D} = \varepsilon \overline{E}$ $\overline{E} = \frac{\overline{D}}{\varepsilon}$ $\overline{E} = \frac{\overline{D}}{\varepsilon} = \frac{Q}{4\pi\varepsilon r^2}\overline{a}_r$ V/m			
			List the salient features and limitations of Gauss's law. ANS: **Salient features and Limitations to of Gauss's law** ▪ The surface may be irregular but should be sufficiently large enough to enclose the charge ▪ The surface must be closed ▪ At each point of surface is either normal or tangential to the surface. ▪ The electric flux density D is constant over the surface at which is normal. **Applications of Gauss's Law** ▪ Used to find and for point charge, line charge, surface charge, infinite line charge, infinite surface charge	1	1	5

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			▪ Used to find and for spherical shell and charged sphere charge distribution ▪ used to find the charge enclosed or the flux passing through the closed surface			
7	**1**	7	Explain any two applications of Gauss's Law with neat diagram. ANS: $\vec{D}$ **due to Spherical Shell of Charge**	1	2	10

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Consider an imaginary spherical shell of radius 'a'.

The charge is uniformly distributed over its surface with a density ρ_S C/m². Let us find $\overline{E}$ at a point P located at a distance r from the centre such that r > a and r ≤ a, using Gauss's law.

The shell is shown in the Fig. 3.14.

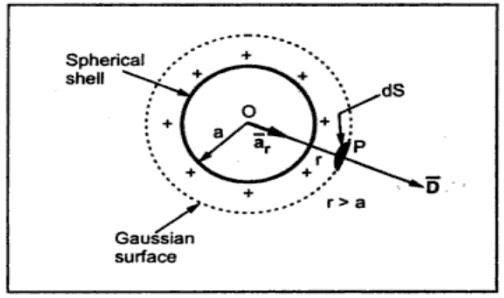

Fig. 3.14 Spherical shell of charge

Case 1 : Point P outside the shell. (r > a)

Consider a point P at a distance r from the origin such that r > a. The Gaussian surface passing through point P is a concentric sphere of radius r. Due to spherical Gaussian surface, the flux lines are directed radially outwards and are normal to the surface. Hence electric flux density $\overline{D}$ is also directed radially outwards at point P and has component only in $\overline{a}_r$ direction. Consider a differential surface area at P normal to $\overline{a}_r$ direction hence $dS = r^2 \sin\theta \, d\theta \, d\phi$ in spherical system.

$$\therefore \quad d\psi = \overline{D} \cdot d\overline{S} = [D_r \overline{a}_r] \cdot \left[r^2 \sin\theta \, d\theta \, d\phi \, \overline{a}_r\right]$$

$$= D_r \, r^2 \sin\theta \, d\theta \, d\phi$$

$$\therefore \quad \psi = \oint_S D_r \, r^2 \sin\theta \, d\theta \, d\phi = D_r \, r^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin\theta \, d\theta \, d\phi$$

$$\therefore \quad \psi = D_r \, r^2 \left[-\cos\theta\right]_0^{\pi} \left[\phi\right]_0^{2\pi} = 4\pi r^2 D_r \qquad \text{... (13)}$$

But $\quad \psi = Q \qquad$ ... Gauss's law

$$\therefore \quad Q = 4\pi r^2 D_r$$

$$\therefore \quad D_r = \frac{Q}{4\pi r^2}$$

$$\therefore \quad \overline{D} = D_r \, \overline{a}_r = \frac{Q}{4\pi r^2} \overline{a}_r \ \text{C/m}^2 \qquad \text{... (14)}$$

And $\quad \overline{E} = \dfrac{\overline{D}}{\varepsilon_0} = \dfrac{Q}{4\pi\varepsilon_0 r^2} \overline{a}_r \ \text{V/m} \qquad$ ... (15)

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Thus for $r > a$, the field $\overline{E}$ is inversely proportional to the square of the distance from the origin.

If the surface charge density is ρ_S C/m² then

$$Q = \rho_S \times \text{Surface area of shell}$$

$$\therefore \quad Q = \rho_S \times 4\pi a^2$$

$$\therefore \quad \overline{E} = \frac{\rho_S 4\pi a^2}{4\pi\varepsilon_0 r^2}\overline{a}_r = \frac{\rho_S a^2}{\varepsilon_0 r^2}\overline{a}_r \text{ V/m} \qquad \text{... (16)}$$

and

$$\boxed{\overline{D} = \varepsilon_0 \overline{E} = \frac{\rho_S a^2}{r^2}\overline{a}_r} \text{ C/m}^2 \qquad \text{... (17)}$$

Case 2 : Point P is on the shell ($r = a$)

On the shell, $r = a$

The Gaussian surface is same as the shell itself and $\overline{E}$ can be obtained using $r = a$ in the equation (15).

$$\therefore \quad \overline{E} = \frac{Q}{4\pi\varepsilon_0 a^2}\overline{a}_r \text{ V/m} \qquad \text{... (18)}$$

Case 3 : Point P inside the shell ($r < a$)

The Gaussian surface, passing through the point P is again a spherical surface with radius $r < a$.

But it can be seen that the entire charge is on the surface and no charge is enclosed by the spherical shell. And when the Gaussian surface is such that no charge is enclosed, irrespective of any charges present outside, the total charge enclosed is zero.

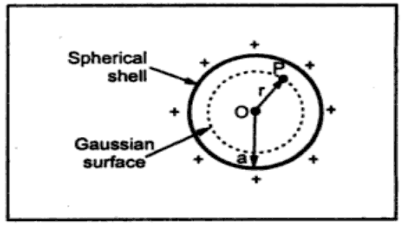

Fig. 3.15

$$\therefore \quad \psi = Q = \oint_S \overline{D} \cdot d\overline{S} = 0 \qquad \text{.... As per Gauss's law}$$

Now $$\oint_S d\overline{S} = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} r^2 \sin\theta \, d\theta \, d\phi = 4\pi r^2$$

Thus $$\oint_S d\overline{S} \neq 0$$

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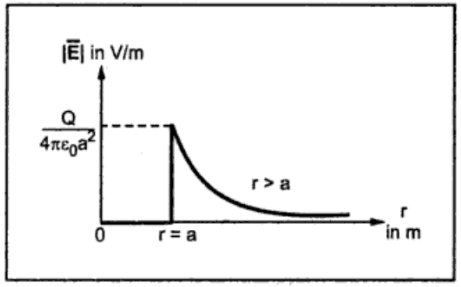
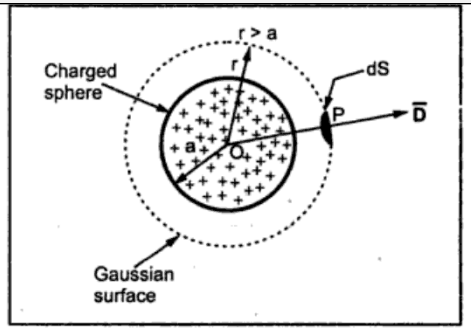
			Hence to satisfy that total charge enclosed is zero, inside the spherical shell. $\bar{D} = 0$ and $\bar{E} = \frac{\bar{D}}{\varepsilon_0} = 0$ ... (19) Thus electric flux density and electric field at any point **inside a spherical shell is zero.** **3.8.5.1 Variation of $\bar{E}$ Against r** The variation of $\bar{E}$ against the radial distance r measured from the origin is shown in the Fig. 3.16. For r < a, $\bar{E} = 0$ For r = a, $\bar{E} = \frac{Q}{4\pi\varepsilon_0 a^2}\bar{a}_r$ For r > a, $\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\bar{a}_r$  Fig. 3.16 Variation of $\|\bar{E}\|$ against r After r = a, the $\bar{E}$ is inversely proportional to the square of the radial distance of a point from the origin. **The variation of $\|\bar{D}\|$ against r is also similar**. For the medium other than the free space, ε_0 must be replaced by $\varepsilon = \varepsilon_0 \varepsilon_r$. **$\vec{D}$ due to Charged Sphere**			

TLO. No:	Module No:	Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	CO	BL	Marks per Q
			Consider a sphere of radius 'a' with a uniform charge density of ρ_v C/m^3. Let us find $\overline{E}$ at a point P located at a radial distance r from centre of the sphere such that $r \le a$ and $r > a$, using Gauss's law. The sphere is shown in the Fig. 3.17. **Case 1 :** The point P is outside the sphere ($r > a$). The Gaussian surface passing through point P is a spherical surface of radius r.  **Fig. 3.17 Uniformly charged sphere** The flux lines and $\overline{D}$ are directed radially outwords along $\overline{a}_r$ direction. The differential area dS is considered at point P which is normal to $\overline{a}_r$ $\therefore \quad dS = r^2 \sin\theta \, d\theta \, d\phi$			

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			$\therefore \quad d\psi = \overline{D} \cdot d\overline{S} = D_r \overline{a}_r \cdot r^2 \sin\theta \, d\theta \, d\phi \, \overline{a}_r$ $= D_r r^2 \sin\theta \, d\theta \, d\phi \quad \ldots (\overline{a}_r \cdot \overline{a}_r = 1)$ $\therefore \quad \psi = Q = \oint_S \overline{D} \cdot d\overline{S} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} D_r r^2 \sin\theta \, d\theta \, d\phi$ $= D_r r^2 [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi} = D_r r^2 4\pi$ $\therefore \quad D_r = \frac{Q}{4\pi r^2}$ $\therefore \quad \overline{D} = \frac{Q}{4\pi r^2} \overline{a}_r \ C/m^2 \quad \ldots (20)$ $\therefore \quad \overline{E} = \frac{\overline{D}}{\varepsilon_0} = \frac{Q}{4\pi\varepsilon_0 r^2} \overline{a}_r \ V/m \quad \ldots (21)$ The total charge enclosed can be obtained as, $Q = \int_v \rho_v dv = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{a} \rho_v r^2 \sin\theta \, dr \, d\theta \, d\phi$ $= \rho_v \left[\frac{r^3}{3}\right]_0^a [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi}$ $= \frac{4}{3}\pi a^3 \rho_v \ C \quad \ldots (22)$ $\therefore \quad \overline{E} = \frac{\frac{4}{3}\pi a^3 \rho_v}{4\pi\varepsilon_0 r^2} \overline{a}_r = \frac{a^3 \rho_v}{3\varepsilon_0 r^2} \overline{a}_r \quad \ldots (23)$ While $\boxed{\overline{D} = \frac{a^3 \rho_v}{3r^2} \overline{a}_r} \quad \ldots (24)$ These are the expressions for $\overline{D}$ and $\overline{E}$ outside the uniformly charged sphere. **Case 2 :** The point P on the sphere (r = a). The Gaussian surface is same as the surface of the charged sphere. Hence results can be obtained directly substituting r = a in the equation (23) and (21). $\therefore \quad \overline{E} = \frac{a^3 \rho_v}{3\varepsilon_0 a^2} \overline{a}_r = \frac{\rho_v a}{3\varepsilon_0} \overline{a}_r \quad \ldots (25)$			

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and $\boxed{\overline{D} = \varepsilon_0 \overline{E} = \frac{\rho_v a}{3} \overline{a}_r}$ ... (26)

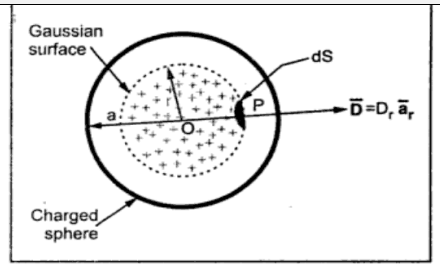

Fig. 3.18

Case 3 : The piont P is inside the sphere (r < a) the Gaussian surface is a spherical surface of radius r where r < a.

Consider differential surface area dS as shown in the Fig. 3.18.

Again $d\overline{S}$ and $\overline{D}$ are directed radially outwards.

$$\therefore \quad \overline{D} = D_r \overline{a}_r \quad \text{while} \quad d\overline{S} = r^2 \sin\theta \, d\theta \, d\phi \, \overline{a}_r$$

$$\therefore \quad d\psi = \overline{D} \cdot d\overline{S} = D_r r^2 \sin\theta \, d\theta \, d\phi \qquad \ldots (\overline{a}_r \cdot \overline{a}_r = 1)$$

$$\therefore \quad \psi = Q = \oint_S \overline{D} \cdot d\overline{S} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} D_r r^2 \sin\theta \, d\theta \, d\phi$$

$$= D_r r^2 [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi} = 4\pi r^2 D_r$$

$$\therefore \quad D_r = \frac{Q}{4\pi r^2}$$

$$\therefore \quad \overline{D} = \frac{Q}{4\pi r^2} \overline{a}_r \ C/m^2 \qquad \ldots (27)$$

Now the charge enclosed is by the sphere of radius r only and not by the entire sphere. **The charge outside the Gaussian surface will not affect $\overline{D}$.**

$$\therefore \quad Q = \int_v \rho_v \, dv = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \frac{4}{3}\pi r^3 \rho_v \qquad \text{where } r < a \qquad \ldots (28)$$

Using in (27) we get,

$$\overline{D} = \frac{\frac{4}{3}\pi r^3 \rho_v}{4\pi r^2} \overline{a}_r$$

$$\therefore \quad \boxed{\overline{D} = \frac{r}{3}\rho_v \overline{a}_r \qquad \ldots\ldots 0 < r \le a} \qquad \ldots (29)$$

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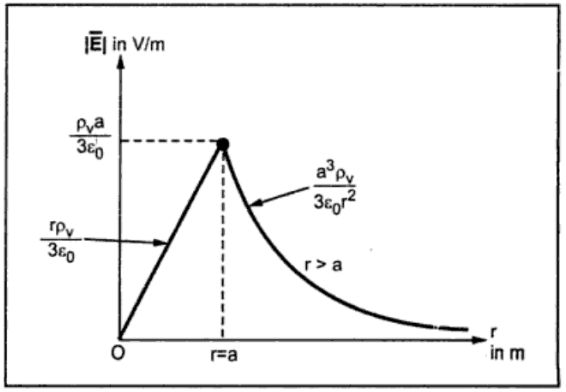
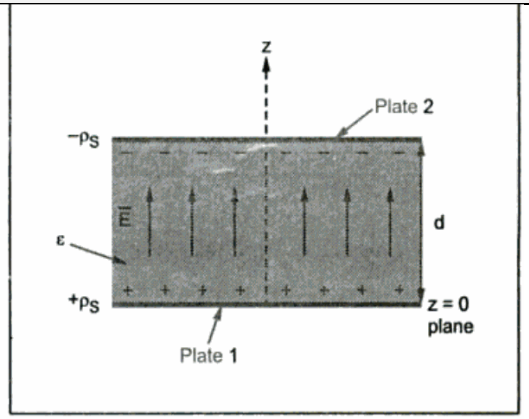
			$\therefore \quad \bar{E} = \frac{\bar{D}}{\varepsilon_0} = \frac{r}{3\varepsilon_0}\rho_v \bar{a}_r \qquad 0 < r \le a \qquad \ldots (30)$ **Key Point**: *The results obtained here can be used as the standard results while solving the problems.* If the sphere is in a medium of permittivity ε_r then ε_0 must be replaced by $\varepsilon = \varepsilon_0 \varepsilon_r$. **3.8.6.1 Variation of $\bar{E}$ Against r** From the equations (21), (23) and (28) it can seen that for r > a, the $\bar{E}$ is inversely proportional to square of the distance while for r < a it is directly proportional to the distance r. At r = a, $\lvert\bar{E}\rvert = \frac{\rho_v a}{3\varepsilon_0}$ depends on the radius of the charged sphere. For r > a, the graph of $\lvert\bar{E}\rvert$ against r is parabolic while for r < a it is a straight line as shown in the Fig. 3.19.  **Fig. 3.19 Variation of $\lvert\bar{E}\rvert$ against r** The graph of $\lvert\bar{D}\rvert$ against r is exactly similar in nature as $\lvert\bar{E}\rvert$ against r.			
8	1	8	Derive an expression for parallel plate capacitance and spherical capacitance. **Parallel Plate Capacitance:**	1	2	10

TLO. No:	Module No:	Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	CO	BL	Marks per Q
			A parallel plate capacitor is shown in the Fig. 5.14. It consists of two parallel metallic plates seperated by distance 'd'. The space between the plates is filled with a dielectric of permittivity ε. The lower plate, plate 1 carries the positive charge and is distributed over it with a charge density $+\rho_S$. The upper plate, plate 2 carries the negative charge and is distributed over its surface with a charge density $-\rho_S$. The plate 1 is placed in z = 0 i.e. xy  Fig. 5.14			

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			plane hence normal to it is z direction. While upper plate 2 is in z = d plane, parallel to xy plane. Let A = Area of cross section of the plates in m^2 $\therefore \quad Q = \rho_S A \; C$ ... (1) This is magnitude of charge on any one plate as charge carried by both is equal in magnitude. To find potential difference, let us obtain $\bar{E}$ between the plates. Assuming plate 1 to be infinite sheet of charge, $\bar{E}_1 = \frac{\rho_S}{2\varepsilon}\bar{a}_N = \frac{\rho_S}{2\varepsilon}\bar{a}_z \;\; V/m$ ... (2) The $\bar{E}_1$ is **normal** at the boundary between conductor and dielectric without any tangential component. While for plate 2, we can write $\bar{E}_2 = \frac{-\rho_S}{2\varepsilon}\bar{a}_N = \frac{-\rho_S}{2\varepsilon}(-\bar{a}_z) \;\; V/m$ ... (3) The direction of $\bar{E}_2$ is downwards i.e. in $-\bar{a}_z$ direction. In between the plates, $\bar{E} = \bar{E}_1 + \bar{E}_2 = \frac{\rho_S}{2\varepsilon}\bar{a}_z + \frac{\rho_S}{2\varepsilon}\bar{a}_z = \frac{\rho_S}{\varepsilon}\bar{a}_z$ ... (4) The potential difference is given by, $V = -\int_{-}^{+} \bar{E}\cdot d\bar{L} = -\int_{upper}^{lower} \frac{\rho_S}{\varepsilon}\bar{a}_z \cdot d\bar{L}$ Now $\overline{dL} = dx\,\bar{a}_x + dy\,\bar{a}_y + dz\,\bar{a}_z$ in cartesian system. $\therefore \quad V = -\int_{z=d}^{z=0} \frac{\rho_S}{\varepsilon}\bar{a}_z \cdot [dx\bar{a}_x + dy\,\bar{a}_y + dz\bar{a}_z]$ $= -\int_{z=d}^{z=0} \frac{\rho_S}{\varepsilon}dz = -\frac{\rho_S}{\varepsilon}[z]_d^0 = -\frac{\rho_S[-d]}{\varepsilon}$ $\therefore \quad V = \frac{\rho_S d}{\varepsilon} \;\; V$ The capacitance is the ratio of charge Q to voltage V,			

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			$C = \frac{Q}{V} = \frac{\rho_S A}{\frac{\rho_S d}{\varepsilon}} = \frac{\varepsilon A}{d}$ F ... (6) Thus if, $\varepsilon = \varepsilon_0 \varepsilon_r$ $\boxed{C = \frac{\varepsilon_0 \varepsilon_r A}{d} \text{ F}}$ ... (7) It can be seen that the value of capacitance depends on, 1. The permittivity of the dielectric used. 2. The area of cross section of the plates. 3. The distance of separation of the plates. It is not dependent on the charge or the potential difference between the plates. **Spherical capacitance:**			

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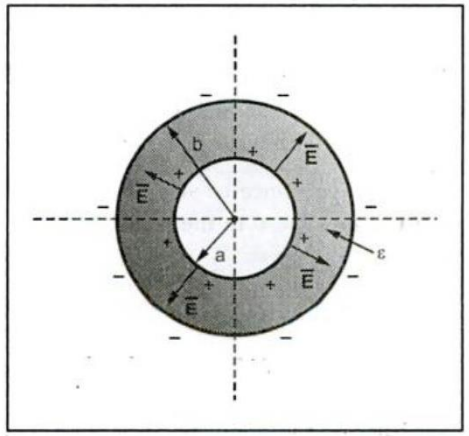
			Consider a spherical capacitor formed of two concentric spherical conducting shells of radius a and b. The capacitor is shown in the Fig. 5.16. The radius of outer sphere is 'b' while that of inner sphere is a'. Thus b > a. The region between the two spheres is filled with a dielectric of permittivity ε. The inner sphere is given a positive charge + Q while for the outer sphere it is – Q. Considering Gaussian surface as a sphere of radius r, it can be obtained that $\overline{E}$ is in radial direction and given by, $\overline{E} = \frac{Q}{4\pi\varepsilon r^2}\overline{a}_r$ V/m ... (1)  Fig. 5.16 Spherical capacitor			

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			The potential difference is work done in moving unit positive dharge **against** the direction of $\bar{E}$ i.e. from r = b to r = a. $\therefore \quad V = -\int_{-}^{+} \bar{E} \cdot d\bar{L} = -\int_{r=b}^{r=a} \frac{Q}{4\pi\varepsilon r^2} \bar{a}_r \cdot d\bar{L}$ ... (2) Now $\quad d\bar{L} = dr\, \bar{a}_r$ ... in radial direction $\therefore \quad V = -\int_{r=b}^{r=a} \frac{Q}{4\pi\varepsilon r^2} \bar{a}_r \cdot dr\, \bar{a}_r = -\int_{r=b}^{r=a} \frac{Q}{4\pi\varepsilon r^2} dr$ $= -\frac{Q}{4\pi\varepsilon}\left[-\frac{1}{r}\right]_{r=b}^{r=a} = \frac{Q}{4\pi\varepsilon}\left[\frac{1}{r}\right]_{r=b}^{r=a}$ $\therefore \quad V = \frac{Q}{4\pi\varepsilon}\left[\frac{1}{a} - \frac{1}{b}\right] V$ ... (3) Now $\quad C = \frac{Q}{V} = -\frac{Q}{\frac{Q}{4\pi\varepsilon}\left[\frac{1}{a} - \frac{1}{b}\right]}$ $\therefore \quad \boxed{C = \frac{4\pi\varepsilon}{\left[\frac{1}{a} - \frac{1}{b}\right]} \text{ F}}$ ... (4)			
9	1	9	Derive the expression for energy density in electrostatic fields in terms of $\vec{D}$ and $\vec{E}$. **Energy density in electrostatic fields in terms of $\vec{D}$ and $\vec{E}$.**	1	2	10

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			Consider the volume charge distribution having uniform charge density ρ_v C / m^3. Hence the total energy stored is given by the equation (10) as, $$W_E = \frac{1}{2}\int_{vol} \rho_v \, V \, dv$$ According to Maxwell's first equation, $$\rho_v = \nabla \cdot \overline{\mathbf{D}}$$			

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			$\therefore \quad W_E = \frac{1}{2}\int_{vol} (\nabla \cdot \overline{D})\ V\ dv$ ... (11) For any vector A and scalar V there is **vector identity,** $\nabla \cdot V\overline{A} = \overline{A} \cdot \nabla V + V(\nabla \cdot \overline{A})$ ... (12) $\therefore \quad (\nabla \cdot \overline{A})\ V = \nabla \cdot V\overline{A} - \overline{A} \cdot \nabla V$ ... (13) Using (13) in (11) we get, $W_E = \frac{1}{2}\int_{vol} (\nabla \cdot V\overline{D} - \overline{D} \cdot \nabla V)\ dv$ $\therefore \quad W_E = \frac{1}{2}\int_{vol} (\nabla \cdot V\overline{D})\ dv - \frac{1}{2}\int_{vol} \overline{D} \cdot \nabla V\ dv$ ... (14) According to divergence theorem, volume integral can be converted to closed surface integral if closed surface totally surrounds the volume. $\therefore \quad \frac{1}{2}\int_{vol} (\nabla \cdot V\overline{D})\ dv = \frac{1}{2}\oint (V\overline{D}) \cdot d\overline{S}$ ... (15) $\therefore \quad W_E = \frac{1}{2}\oint (V\overline{D}) \cdot d\overline{S} - \frac{1}{2}\int_{vol} \overline{D} \cdot \nabla V\ dv$ ... (16) We know that $V \propto \frac{1}{r}$ and $\overline{D} \propto \frac{1}{r^2}$ for point charge, $V \propto \frac{1}{r^2}$, $\overline{D} \propto \frac{1}{r^3}$ for dipoles and so on. So $V\overline{D}$ is proportional to at least $1/r^3$ while dS varies as r^2. Hence total integral varies as $1/r$. As surface becomes very large, $r \to \infty$ and $1/r \to 0$. Hence **closed surface integral is zero** in the equation (16). $\therefore \quad W_E = -\frac{1}{2}\int_{vol} \overline{D} \cdot \nabla V\ dv$ ... (17) But $\quad \overline{E} = -\nabla V$ $\therefore \quad W_E = -\frac{1}{2}\int_{vol} \overline{D} \cdot (-\overline{E})\ dv$ ... (18) $\therefore \quad \boxed{W_E = \frac{1}{2}\int_{vol} \overline{D} \cdot \overline{E}\ dv\ \text{J}}$ ... (19) Now $\quad \overline{D} = \varepsilon_0 \overline{E}$			

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			$\therefore \quad W_E = \frac{1}{2}\int_{vol} \varepsilon_0 \bar{E}\cdot\bar{E}\ dv\ J$ $\therefore \quad \boxed{W_E = \frac{1}{2}\int_{vol} \varepsilon_0 E^2\ dv\ J}$ ... as $\bar{E}\cdot\bar{E} = E^2$ ... (20) $\therefore \quad \boxed{W_E = \frac{1}{2}\int_{vol} \frac{D^2}{\varepsilon_0}\ dv\ J}$ ... (21) In a differential form, $d W_E = \frac{1}{2}\bar{D}\cdot\bar{E}\ dv$ $\therefore \quad \boxed{\frac{d W_E}{dv} = \frac{1}{2}\bar{D}\cdot\bar{E}\ J/m^3}$ ... (22) This is called **energy density** in the electric field having units J/m^3. If this is integrated over the volume, we get total energy present. $\boxed{W_E = \int_{vol}\left(\frac{d W_E}{dv}\right) dv}$ ... (23)			
10	1	10	Derive Continuity current equation for electrostatics. **Continuity current equation:**	1	2	10

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			The continuity equation of the current is based on the **principle of conservation of charge**. The principle states that, The charges can neither be created nor be destroyed. Consider a closed surface S with a current density $\bar{J}$, then the total current I crossing the surface S is given by, $$\boxed{I = \oint_S \bar{J} \cdot d\bar{S}} \quad \text{... (1)}$$ The current flows outwards from the closed surface. It has been mentioned earlier that **the current means the flow of positive charges.** Hence the current I is constituted due to outward flow of positive charges from the closed surface S. According to principle of conservation of charge, there must be decrease of an equal amount of positive charge inside the closed surface. Hence the outward rate of flow of positive charge gets balanced by the rate of decrease of charge inside the closed surface. Let Q_i = Charge within the closed surface $-\frac{dQ_i}{dt}$ = Rate of **decrease** of charge inside the closed surface The negative sign indicates decrease in charge.			

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			Due to principle of conservation of charge, this rate of decrease is same as rate of outward flow of charge, which is a current. $\therefore \quad I = \oint_S \bar{J} \cdot d\bar{S} = -\frac{dQ_i}{dt}$ ... (2) This is the **integral form of the continuity equation of the current.** The negative sign in the equation indicates outward flow of current from the closed surface. So the equation (2) is indicating **outward flowing current I.** If the current is entering the volume then $\oint_S \bar{J} \cdot d\bar{S} = -I = +\frac{dQ_i}{dt}$ The point form of the continuity equation can be obtained from the integral form. Using the **divergence theorem**, convert the surface integral in integral form to the volume integral. $\therefore \quad \boxed{\oint_S \bar{J} \cdot d\bar{S} = \int_{vol} (\nabla \cdot \bar{J})\, dv}$ ...(3) $\therefore \quad -\frac{dQ_i}{dt} = \int_{vol} (\nabla \cdot \bar{J})\, dv$ ... (4) But $\quad Q_i = \int_{vol} \rho_v \, dv$ ... (5) where $\quad \rho_v$ = Volume charge density $\therefore \quad \int_{vol} (\nabla \cdot \bar{J})\, dv = -\frac{d}{dt}\left[\int_{vol} \rho_v \, dv\right] = -\int_{vol} \frac{\partial \rho_v}{\partial t}\, dv$ ... (6) For a constant surface, the derivative becomes the partial derivative. $\therefore \quad \int_{vol} (\nabla \cdot \bar{J})\, dv = \int_{vol} -\frac{\partial \rho_v}{\partial t}\, dv$ ... (7) If the relation is true for any volume, it must be true even for incremental volume Δv. $\therefore \quad (\nabla \cdot \bar{J})\Delta v = -\frac{\partial \rho_v}{\partial t}\Delta v$ ... (8)			

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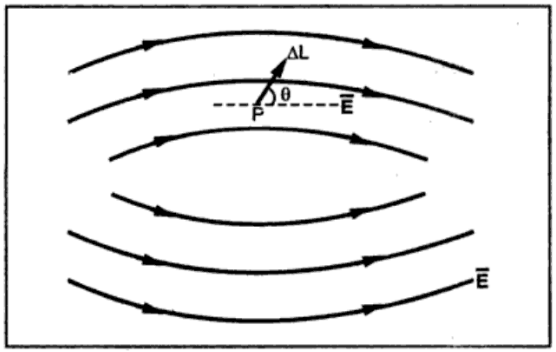
			$\therefore \quad \boxed{\nabla \cdot \bar{J} = -\frac{\partial \rho_v}{\partial t}}$ ... (9) This is the **point form or differential form of the continuity equation of the current.** The equation states that the current or the charge per second, diverging from a small volume per unit volume is equal to the time rate of decrease of charge per unit volume at every point. **5.3.1 Steady Current** For steady currents which are not the functions of time, $\partial \rho_v / \partial t = 0$ hence, $\nabla \cdot \bar{J} = 0$ (Steady current) ... (10) For such currents, the rate of flow of charge remains constant with time. The steady currents have no sources or sinks, as it is constant.			
11	1	11	Define potential gradient and derive the relationship between E and V. **Relationship between E and V** Consider $\bar{E}$ due to a particular charge distribution in space. The electric field $\bar{E}$ and potential V is changing from point to point in space. Consider a vector incremental length $\Delta\bar{L}$ making an angle θ with respect to the direction of $\bar{E}$, as shown in the Fig. 4.29.  Fig. 4.29 Incremental length at an angle θ To find incremental potential we use,	1	2	10

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			$\Delta V = -\bar{E} \cdot \Delta \bar{L}$ ... (1) Now $\Delta \bar{L} = \Delta L\, \bar{a}_L$ ... (2) where $\bar{a}_L$ = Unit vector in the direction of ΔL. Now the dot product means product of magnitudes of one quantity and component of other in the direction of first. So $\bar{E} \cdot \Delta \bar{L}$ is the product of component of $\bar{E}$ in the direction of $\bar{a}_L$ and ΔL. $\therefore \quad \bar{E} \cdot \Delta \bar{L} = (E_L\, \bar{a}_L) \cdot (\Delta L\, \bar{a}_L)$ ... $\bar{a}_L \cdot \bar{a}_L = 1$ $\therefore \quad \bar{E} \cdot \Delta \bar{L} = E_L\, \Delta L$ $\therefore \quad \Delta V = -E_L\, \Delta L$ ... (3) where E_L = Component of $\bar{E}$ in the direction of $\bar{a}_L$. In other words, dot product can be expressed in terms of cos θ as. $\Delta V = -E\, \Delta L \cos\theta$ ... as $\bar{E} \cdot \Delta \bar{L} = \lvert E \rvert \lvert \Delta L \rvert \cos\theta$ $\therefore \quad \frac{\Delta V}{\Delta L} = -E \cos\theta$ To find ΔV at a point, take $\lim \Delta L \to 0$, $\therefore \quad \lim_{\Delta L \to 0} \frac{\Delta V}{\Delta L} = -E \cos\theta$ ... (5) But $\lim_{\Delta L \to} \frac{\Delta V}{\Delta L} = \frac{dV}{dL}$ = Potential gradient $\therefore \quad \frac{dV}{dL} = -E \cos\theta$ ... (6) At a point P where ΔL is considered, $\bar{E}$ has a fixed value while ΔL is also constant. Hence potential gradient $\frac{dV}{dL}$ can be maximum only when $\cos\theta = -1$ i.e. $\theta = +180°$. **This indicates that ΔL must be in the direction opposite to $\bar{E}$.** $\therefore \quad \left.\frac{dV}{dL}\right\rvert_{max} = E$ ... (7) This equation shows that, 1. Maximum value of the potential gradient gives the magnitude of the electric field intensity $\bar{E}$.			

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			2. The maximum value of rate of change of potential with distance i.e. potential gradient is possible only when the direction of increment in distance is opposite to the direction of $\overline{E}$. Thus if $\overline{a}_n$ is the unit vector in the direction of **increasing potential** normal to the equipotential surface then $\overline{E}$ can be expressed as, $$\overline{E} = -\left.\frac{dV}{dL}\right\|_{max} \overline{a}_n \quad \text{... (8)}$$ As $\overline{E}$ and potential gradient are in opposite direction, equation (8) has a negative sign. The equation shows that the magnitude of $\overline{E}$ is given by maximum space rate of change of V while the direction of $\overline{E}$ is normal to the equipotential surface in the direction of **decreasing potential.** The **maximum value of rate of change of potential with distance (dV/dL) is called gradient of V.** The mathematical operation on V by which $-\overline{E}$ is obtained is called **gradient** and denoted as, Gradient of V = grad V = ∇ V ... (9) $\therefore$ $\nabla V = \text{grad } V = -\overline{E}$ V/m ... (10) or $\boxed{\overline{E} = -\nabla V = -(\text{grad } V)}$ ... (11) The equation (11) gives the relationship between $\overline{E}$ and V. Now $\overline{E}$ is vector but V is scalar, hence remember that grad V i.e. **gradient of a scalar is a vector.** **Another Method:**			

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			In space the potential V is unique function of x, y and z coordinates, in cartensian system denoted as V (x, y, z). Hence its total differential potential dV can be obtained as, $$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz \quad \text{... (12)}$$ In cartesian coordinates, $$\bar{E} = E_x \bar{a}_x + E_y \bar{a}_y + E_z \bar{a}_z \quad \text{... (13)}$$ While $$d\bar{L} = dx\,\bar{a}_x + dy\,\bar{a}_y + dz\,\bar{a}_z \quad \text{... (14)}$$ $$\therefore \quad dV = -\bar{E} \cdot d\bar{L} = -[E_x\,dx + E_y\,dy + E_z\,dz] \quad \text{... (15)}$$ Comparing equations (12) and (15) we can write, $$E_x = -\frac{\partial V}{\partial x},\ E_y = -\frac{\partial V}{\partial y},\ E_z = -\frac{\partial V}{\partial z} \quad \text{... (16)}$$ Hence $\bar{E}$ can be expressed interms of (16) as, $$\bar{E} = -\frac{\partial V}{\partial x}\bar{a}_x - \frac{\partial V}{\partial y}\bar{a}_y - \frac{\partial V}{\partial z}\bar{a}_z$$ $$\therefore \quad \bar{E} = -\left[\frac{\partial}{\partial x}\bar{a}_x + \frac{\partial}{\partial y}\bar{a}_y + \frac{\partial}{\partial z}\bar{a}_z\right] V \quad \text{... (17)}$$ The potential V is scalar but the operator on V given in equation is vector and is called **del** operator denoted as ∇. The operation of del on a scalar V is called grad V. $$\therefore \quad \nabla = \frac{\partial}{\partial x}\bar{a}_x + \frac{\partial}{\partial y}\bar{a}_y + \frac{\partial}{\partial z}\bar{a}_z \quad \text{... (18)}$$ $$\therefore \quad \boxed{\bar{E} = -\nabla V} \quad \text{... (19)}$$ **The grad of a scalar is a vector.**			
12	1	12	Derive the expression for total dipole moment in a dielectric medium.	1	2	10

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			5.6.2 Mathematical Expression for Polarization When the dipole is formed due to polarization, there exists an electric dipole moment $\overline{p}$. $$\boxed{\overline{p} = Q\,\overline{d}} \quad \text{... (1)}$$ where Q = Magnitude of one of the two charges $\overline{d}$ = Distance vector from negative to positive charge Let n = Number of dipoles per unit volume Δv = Total volume of the dielectric N = Total dipoles = $n\,\Delta v$ Then the total dipole moment is to be obtained using superposition principle as, $$\overline{p}_{total} = Q_1\overline{d}_1 + Q_2\overline{d}_2 + \ldots + Q_n\overline{d}_n = \sum_{i=1}^{n\Delta v} Q_i\,\overline{d}_i \quad \text{... (2)}$$ If dipoles are randomly oriented, $\overline{p}_{total}$ is zero but if dipoles are aligned in the direction of applied $\overline{E}$ then $\overline{p}_{total}$ has a significant value. The **polarization $\overline{P}$ is defined as the total dipole moment per unit volume**. $$\therefore \quad \boxed{\overline{P} = \lim_{\Delta v \to 0} \frac{\sum_{i=1}^{n\Delta v} Q_i\,\overline{d}_i}{\Delta v}} \quad \text{... (3)}$$ It is measured in coulombs per square metre (C/m^2).			